

How much does shape explain?



Flow Lenia specimen 870da94a. Native body rendering from a 256×256 simulation grid. This specimen belongs to the separate obstacle experiment, not the fish-near group.

How much does shape explain?

Comparing Flow Lenia, simulated tissue, and fish

We measured the shapes of 25,167 patterns generated by Flow Lenia, then compared them with measurements of real fish. A group chosen before looking at the fish data had unexpectedly similar proportions.

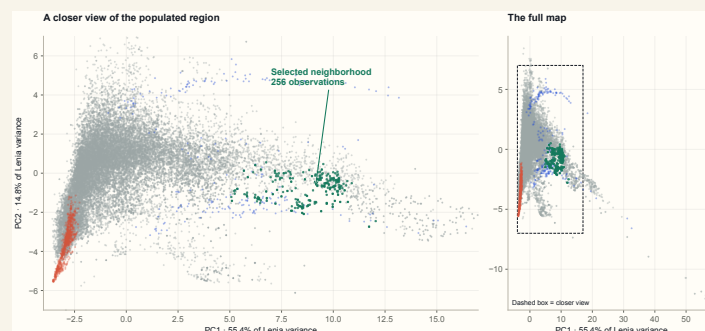
The comparison uses twelve normalized measurements. The selected group contains 256 observations. Its median distance to the nearest fish is 3.30, compared with 5.90 across the full Lenia sample.

These are distances in a space of measurements. They do not mean that the simulated patterns have fish anatomy, or that the same process produced them. The next question is whether similar shapes respond similarly when disturbed.

The report follows that question from the shape map to a separate experiment with moving bodies and obstacles.

5.90 → 3.30

Median distance to the nearest fish: full sample → selected group.



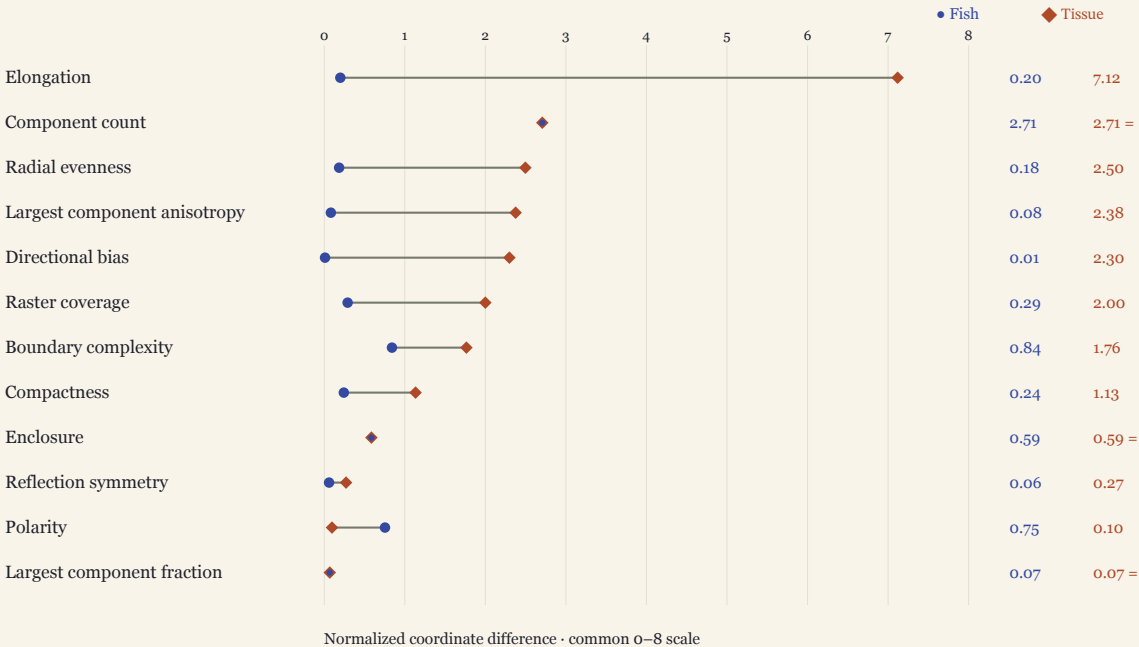
Full investigation and data

Layout study, September 2026. This excerpt is a publication prototype, not the complete research article.

Comparison plate

Where the resemblance holds.

flow-map-elite-53 and its nearest fish and tissue neighbors



= Exact agreement between the two differences, not a missing mark.

Four fish, eleven landmarks each.

Same x–y projection and coordinate scale; points are measured locations, not body outlines.



Source: report figure data and the cited fish landmark study. Positions retain their published normalization.

Figure sources

The comparison plate is generated from the report's published fish-explainer.json. Each row compares one Lenia pattern, flow-map-elite-53, with its nearest fish and simulated-tissue neighbors. All twelve coordinate differences share one scale.

The cover specimen, when shown, comes from the separate obstacle experiment. It is not evidence for the fish-near comparison.

Fish landmark study · Scientific Reports

Visual references · Phil Rohr's Are.na collection

Format references: image-led editorial pages, Synergetics of Cognition, and the Bell System Technical Journal. Historical references inform composition and typography; the figures in this prototype come from the research.